In earlier classes, using the above relations between roots and coefficients we constructed a quadratic equation, having $\alpha$ and $\beta$ as roots. In fact, such an equation is given by (1). For instance, a quadratic equation whose roots are 3 and 4 is given by $x^2 - 7x + 12 = 0$.

Further we construct new polynomial equations whose roots are functions of the roots of a given polynomial equation; in this process we form a new polynomial equation without finding the roots of the given polynomial equation. For instance, we construct a polynomial equation by increasing the roots of a given polynomial equation by two as given below.

**Example 3.1**

If $\alpha$ and $\beta$ are the roots of the quadratic equation $17x^2 + 43x - 73 = 0$, construct a quadratic equation whose roots are $\alpha + 2$ and $\beta + 2$.

**Solution**

Since $\alpha$ and $\beta$ are the roots of $17x^2 + 43x - 73 = 0$, we have $\alpha + \beta = \frac{-43}{17}$ and $\alpha\beta = \frac{-73}{17}$.

We wish to construct a quadratic equation with roots $\alpha + 2$ and $\beta + 2$. Thus, to construct such a quadratic equation, calculate,

$$\text{the sum of the roots} = \alpha + \beta + 4 = \frac{-43}{17} + 4 = \frac{25}{17} \quad \text{and}$$

$$\text{the product of the roots} = \alpha\beta + 2(\alpha + \beta) + 4 = \frac{-73}{17} + 2\left(\frac{-43}{17}\right) + 4 = \frac{-91}{17}.$$

Hence a quadratic equation with required roots is $x^2 - \frac{25}{17}x - \frac{91}{17} = 0$.

Multiplying this equation by 17, gives $17x^2 - 25x - 91 = 0$

which is also a quadratic equation having roots $\alpha + 2$ and $\beta + 2$.

**Example 3.2**

If $\alpha$ and $\beta$ are the roots of the quadratic equation $2x^2 - 7x + 13 = 0$, construct a quadratic equation whose roots are $\alpha^2$ and $\beta^2$.

**Solution**

Since $\alpha$ and $\beta$ are the roots of the quadratic equation, we have $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = \frac{13}{2}$.

Thus, to construct a new quadratic equation,

$$\text{Sum of the roots} = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{-3}{4}.$$

$$\text{Product of the roots} = \alpha^2\beta^2 = (\alpha\beta)^2 = \frac{169}{4}$$

Thus a required quadratic equation is $x^2 + \frac{3}{4}x + \frac{169}{4} = 0$. From this we see that

$$4x^2 + 3x + 169 = 0$$

is a quadratic equation with roots $\alpha^2$ and $\beta^2$.

**Remark**

In Examples 3.1 and 3.2, we have computed the sum and the product of the roots using the known $\alpha + \beta$ and $\alpha\beta$. In this way we can construct quadratic equation with desired roots, provided the sum and the product of the roots of a new quadratic equation can be written using the sum and the product of the roots of the given quadratic equation. We note that we have not solved the given equation; we do not know the values of $\alpha$ and $\beta$ even after completing the task.



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